

In-class exercise 4 (solution)

Compute the interplanar spacing in a tetragonal crystal

$$\vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l][\Gamma^*] \begin{bmatrix} h \\ k \\ l \end{bmatrix} \quad \Gamma^* = \begin{bmatrix} a^{*2} & a^*b^*\cos\gamma^* & a^*c^*\cos\beta^* \\ a^*b^*\cos\gamma^* & b^{*2} & b^*c^*\cos\alpha^* \\ a^*c^*\cos\beta^* & b^*c^*\cos\alpha^* & c^{*2} \end{bmatrix}$$

$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{V} = \frac{\vec{b} \times \vec{c}}{\vec{a} \cdot (\vec{b} \times \vec{c})}$$

$$a=b \neq c; \alpha=\beta=\gamma=90^\circ$$

$$\vec{a} = a\hat{x} \quad \vec{b} = a\hat{y} \quad \vec{c} = c\hat{z}$$

$$a^*=b^* \neq c^*; \alpha^*=\beta^*=\gamma^*=90^\circ$$

$$V = \begin{vmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & c \end{vmatrix} = a^2c$$

$$\Gamma_{tetragonal}^* = \begin{bmatrix} a^{*2} & 0 & 0 \\ 0 & a^{*2} & 0 \\ 0 & 0 & c^{*2} \end{bmatrix}$$

$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{a^2c} = \frac{1}{a^2c} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & a & 0 \\ 0 & 0 & c \end{vmatrix} = \frac{ac}{a^2c} \hat{x} = \frac{1}{a} \hat{x}$$

$$\vec{b}^* = \frac{\vec{c} \times \vec{a}}{a^2c} = \frac{1}{a^2c} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & c \\ a & 0 & 0 \end{vmatrix} = \frac{ac}{a^2c} \hat{y} = \frac{1}{a} \hat{y}$$

$$\vec{c}^* = \frac{\vec{a} \times \vec{b}}{a^2c} = \frac{1}{a^2c} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ a & 0 & 0 \\ 0 & a & 0 \end{vmatrix} = \frac{a^2}{a^2c} \hat{z} = \frac{1}{c} \hat{z}$$

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$$\Gamma_{tetragonal}^* = \begin{bmatrix} \frac{1}{a^2} & 0 & 0 \\ 0 & \frac{1}{a^2} & 0 \\ 0 & 0 & \frac{1}{c^2} \end{bmatrix} \quad \vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l][\Gamma^*] \begin{bmatrix} h \\ k \\ l \end{bmatrix}$$

$$\frac{1}{d_{hkl}^2} = [h \quad k \quad l] \begin{bmatrix} \frac{1}{a^2} & 0 & 0 \\ 0 & \frac{1}{a^2} & 0 \\ 0 & 0 & \frac{1}{c^2} \end{bmatrix} \begin{bmatrix} h \\ k \\ l \end{bmatrix} = [h \quad k \quad l] \begin{bmatrix} \frac{h}{a^2} \\ \frac{k}{a^2} \\ \frac{l}{c^2} \end{bmatrix} = \frac{h^2}{a^2} + \frac{k^2}{a^2} + \frac{l^2}{c^2}$$

$$d_{hkl} = \frac{1}{\sqrt{\frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}}}$$